



TERRASATCH

TerraSatch V1 Whitepaper: Terrain Intelligence for Safer Decisions

Version v1.3 - March 05, 2026

Prepared for avalanche and mountain safety organizations, land managers, infrastructure and utility operators, ski areas, SAR teams, municipalities, insurers, outdoor brands, and technical partners.

Website: <https://www.terrasatch.com>



0. Front Matter

0.1 Title Page (version, date)

Document Title: TerraSatch V1 Whitepaper: Terrain Intelligence for Safer Decisions

Version: v1.3

Date: March 05, 2026

0.2 Disclaimer (safety/legal)

TerraSatch provides decision-support software intended to improve terrain awareness and operational coordination. TerraSatch does not prevent accidents and does not replace professional training, certified avalanche education, local advisories, engineering judgment, or organizational safety protocols.

Avalanche terrain, weather, snowpack, wildfire behavior, hydrology, and slope stability involve uncertainty, incomplete information, and rapidly changing conditions. Users must verify outputs against field observations, official forecasts and warnings, and site-specific expertise.

This whitepaper describes V1 scope and near-term roadmap. Features described as future are not commitments unless explicitly contracted in a statement of work.

0.3 Contact / Links

Website: <https://www.terrasatch.com>

Data: <https://data.terrasatch.com>



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1. Executive Summary

TerraSatch is a terrain intelligence platform born in the Wasatch and built for field teams. It fuses drone flights, LiDAR and terrain models, weather and advisory inputs, and standardized field observations into operational maps, alerts, and reports.

V1 delivers a ship-ready foundation: TerraGrid (the base terrain layer), AvyTS (avalanche and snowpack intelligence) as the flagship module, shared multi-hazard module scaffolding, offline-first field workflows, evidence-linked reporting, and API-first integration (REST and GraphQL). V1 is decision support. Outputs are reviewable, versioned, and auditable.

1.1 Assumptions used in this document

This document mirrors TerraSatch.com for branding, module naming, and stated platform capabilities. Where TerraSatch.com presents high-level concepts (e.g., “blockchain verified” or “quantum ready”), this whitepaper describes V1 implementation as an auditable data pipeline with optional advanced components that can be enabled by deployment and partner requirements.

1.2 V1 scope boundaries (in scope, next, out of scope)

In scope for V1: TerraGrid base terrain layers and derivatives; AvyTS V1 overlays and observation workflows; shared Terrain Graph with lineage/versioning and audit logs; offline-first mobile capture and sync; web map UX (layers + time slider + reporting); documented core APIs and GIS-friendly exports.

Next (post-V1, near-term): deeper cross-module coupling (AvyTS ↔ HydroTS); expanded GeoTS corridor pilots; additional automation with calibration and uncertainty; enterprise integration templates and webhooks.

Explicitly out of scope for V1: automated go/no-go decisions; guarantees of safety or incident prevention; engineering-stamped certification products; mandatory hardware/mesh/LEO deployments; black-box models without audit trails; universal, fully calibrated models across all regions without partner-specific validation.

2. Problem Statement

2.1 The gap in current tools/workflows

Terrain decisions are often made with fragmented, outdated, and hard-to-use data. Hazard context lives across separate tools: GIS desktops, advisories, weather products, drone deliverables, radio comms, and local notes. During critical windows, teams waste time reconciling these sources, miss signals, and struggle to produce consistent briefings and defensible incident packages.

2.2 Why timing + terrain complexity breaks existing approaches

Terrain hazards are time-dependent. Snowpack can change within hours; wildfire risk shifts with wind and fuels; hydrology responds to precipitation and melt; rockfall and slope stability evolve with freeze–thaw and disturbances. When conditions change faster than teams can update their shared operational picture, decisions degrade. V1 focuses on reducing that gap with a consistent terrain



model, rapid updates, offline capture, and evidence-linked outputs.

3. TerraSatch Overview

3.1 Mission + vision

Mission: deliver terrain intelligence that field teams can actually use—day-to-day, on-device, and under pressure.

Vision: a unified terrain intelligence layer across the West that scales across hazards while remaining integration-friendly with existing GIS and partner systems.

3.2 Core product thesis: Terrain Intelligence

TerraSatch organizes “terrain intelligence” as a layered system:

- TerraGrid: unified terrain model from LiDAR, satellite DEMs, and drone orthomosaics (foundation for all analysis).
- Module layers: AvyTS, GeoTS, HydroTS, PyroTS, InfraTS share a common terrain foundation and data pipeline.
- Ops layer: routes, tasks, field observations, media, and decision logs.
- Signal layer: alerts, watch zones, and published briefings tied to snapshots and reviewer approval.

3.3 Who it's for (primary users + secondary users)

Primary users: ski patrol and mountain ops teams; avalanche forecasters; backcountry guides; SAR and incident leads; field technicians inspecting terrain-adjacent assets.

Secondary users: land managers and municipalities; transportation corridor operators; utilities and critical infrastructure operators; insurers and risk analysts; technical partners integrating TerraSatch into dashboards and GIS stacks.

3.4 Platform metrics (reported on TerraSatch.com)

The following platform metrics are presented on TerraSatch.com as current indicators of scale and usage:

- 10,000+ square miles mapped (Western U.S. coverage)
- 50+ field partners
- 720+ daily forecasts (AvyTS-powered predictions)
- 2.3M+ API calls/day (real-time data access)
- 99.9% field uptime

4. System Modules (V1)

V1 delivers a shared terrain foundation (TerraGrid) and a common Terrain Graph that ties together terrain derivatives, observations, module outputs, and decision logs with lineage metadata. AvyTS is



the flagship research pilot and V1 operational module; the other modules ship with foundational layers and targeted analytics suitable for pilots and screening use cases.

4.1 AvyTS (AvyTerraSatch) — Avalanche / Snowpack

a) What it does

Avalanche and snowpack intelligence layer that connects terrain, weather/advisory inputs, and field observations to support stability interpretation, avalanche-path awareness, and operational briefings.

b) Inputs (data sources)

TerraGrid terrain derivatives; LiDAR/DEM surfaces; drone orthomosaics; weather forecasts and station networks; public advisories/archives where available; standardized field observations (routes, pits, photos, notes).

c) Outputs (maps, alerts, reports, APIs)

Hazard overlays by AOI and time window; slope/aspect/elevation band summaries; storm-cycle timeline views; weak-layer tracking; shareable briefings and incident packages; AvyTS APIs.

d) V1 capabilities vs future

V1: Operational overlays with explicit uncertainty; observation schema and review workflow; reproducible reports (snapshot + versions); offline-first capture with sync.

Future: Higher-fidelity ensembles with calibration, expanded route optimization support, and increased automation guarded by auditability and validation.

4.2 GeoTS — Geology / Rockfall / Slope Stability

a) What it does

Geology and rockfall intelligence for screening-level slope stability, corridor prioritization, and evidence packaging. GeoTS is not an engineering certification product.

b) Inputs (data sources)

LiDAR/DEMs and drone photogrammetry; common public geologic datasets; partner-provided inventories and corridor context (by agreement); field observations (rockfall evidence, crack mapping, inspection notes).

c) Outputs (maps, alerts, reports, APIs)

Rockfall susceptibility overlays; slope-stability screening zones; corridor inspection prioritization reports; GIS exports and API layers.

d) V1 capabilities vs future

V1: Operational screening maps with transparent assumptions; event inventories and evidence packages; reviewer-gated publishing.

Future: Partner-calibrated models for priority corridors and higher-fidelity simulation hooks when validated against inventories.

4.3 HydroTS — Hydrology / Flood / Watershed

a) What it does

Watershed and runoff intelligence for event awareness and planning, connecting terrain, precipitation, temperature, and telemetry where available.



b) Inputs (data sources)

Flow direction/accumulation derivatives; watershed boundaries; precipitation/temperature forecasts; gauges and partner telemetry (by agreement); remote sensing indices where available.

c) Outputs (maps, alerts, reports, APIs)

Runoff and flood watch overlays; basin summaries; exposure views for corridors and assets; event reports and exports.

d) V1 capabilities vs future

V1: Event-based awareness products with explicit assumptions; basin summaries and watch zones; pilot calibration where telemetry is available.

Future: Improved coupling to snowpack dynamics and expanded telemetry ingestion with automated watch/alert rules.

4.4 PyroTS — Wildfire / Fuels / Risk

a) What it does

Wildfire fuels and risk context by fusing land cover/fuels, terrain exposure, and weather inputs. PyroTS supports planning and situational awareness and is not a replacement for incident command.

b) Inputs (data sources)

Land cover and fuels datasets; multispectral indices; terrain derivatives; wind/humidity forecasts; burn perimeters and severity datasets where available; field notes and imagery.

c) Outputs (maps, alerts, reports, APIs)

Fire risk overlays and time sliders; burn-scar mapping; change detection layers; incident planning exports; risk summaries for zones and assets.

d) V1 capabilities vs future

V1: Risk context and mapping suitable for planning with clear uncertainty; rapid burn mapping packages.

Future: Higher-frequency satellite fusion and real-time perimeter ingestion where data is available and validated.

4.5 InfraTS — Infrastructure & Environmental

a) What it does

Asset monitoring, inspection workflows, and environmental compliance scoring for roads, utilities, and sensitive sites. InfraTS produces evidence packages and prioritization, not engineering certification.

b) Inputs (data sources)

Partner-provided asset inventories (by agreement); module hazard layers; drone imagery and LiDAR point clouds; field inspections and checklists; telemetry where available.

c) Outputs (maps, alerts, reports, APIs)

Segment/site overlays; inspection task lists; compliance and evidence packages; GIS exports and APIs; enterprise webhooks.

d) V1 capabilities vs future

V1: Inspection workflow + risk overlays + reporting with configurable rubrics and audit logs.

Future: Automated defect detection at scale and deeper integration into work-order systems where



contracted.

4.6 Cross-Module Terrain Graph (shared terrain model)

The Terrain Graph is the shared, versioned representation connecting terrain tiles, derived features, observations, module outputs, events (storms, perimeters), and optional assets. V1 requires that published layers and reports are reproducible from input datasets and pipeline versions with an audit trail of what changed, when, and why.

4.7 SherpAI Agent (TerraSatch field + ops copilot)

SherpAI is TerraSatch's operational AI agent designed to help field teams and ops leads navigate the platform, generate briefings, and translate observations into consistent, structured records. SherpAI is constrained by the same safety principles as the platform: it provides decision support, surfaces evidence, and records rationale. It does not issue go/no-go decisions.

- V1 capabilities: guided observation entry (schema assistance), briefing draft generation from selected evidence, and "what changed" summaries across publish snapshots.
- Enterprise options: policy-aware redaction, role-based response constraints, and audit logging of agent-assisted outputs.

5. Product Experience (V1)

5.1 Web app UX (maps, layers, time slider, reporting)

V1 centers on an interactive map designed for field operators, not only analysts. Core UI elements include layer controls (TerraGrid + module overlays), a time slider for seasonal and event windows, zone summaries, and a reporting panel that generates briefing packets and incident packages.

5.2 Mobile field workflow (offline-first, capture, sync)

V1 mobile is offline-first: users pre-load zones and tiles, capture observations locally with integrity metadata, and synchronize when connectivity is available. Sync is designed to be resilient (resumable uploads, conflict-safe merges, audit logs).

5.3 Reporting outputs (PDF-style reports, share links)

V1 outputs include daily briefings, incident packages, and share links that recreate a specific map view at a specific time and version. Share links are permissioned and preserve snapshot provenance (layers, versions, and included observations).

5.4 Roles & permissions (org admin, forecaster, field tech)

- Org Admin: users, zones, data sharing, API keys, retention settings.
- Forecaster/Analyst: approve observations, publish layers, configure watch thresholds, author briefings.
- Field Tech/Operator: capture observations, view assigned layers, generate reports.
- External Viewer (optional): read-only access to shared briefings (scoped and expiring).



6. Data & Intelligence Pipeline

6.1 Data ingestion (public + partner + field collected)

V1 ingests standard geospatial formats (raster, vector, point clouds), weather feeds, and structured field observations. Ingestion includes spatial validation, timestamp normalization, and integrity checks for auditability.

6.2 Normalization + QA/QC

V1 reduces garbage-in/garbage-out through projection and resolution checks, schema validation, automated quality flags, and human review gates for high-impact publishing.

6.3 Terrain features & derivatives (slope/aspect/curvature/etc.)

TerraGrid derivatives are computed consistently across tiles so cross-module outputs share comparable semantics (e.g., slope, aspect, curvature, roughness, flow accumulation).

6.4 Modeling approach (statistical + ML + rules-based where needed)

V1 prioritizes credibility-first modeling: rules-based logic where domain knowledge is strong; statistical scoring where calibration is feasible; and ML for interpretable intermediate products (segmentation, anomaly candidates). Every output links to input datasets, model/version, and review status.

6.5 Human-in-the-loop review and auditability

V1 distinguishes draft vs published layers. Organizations can require reviewer sign-off for published overlays and high-severity alerts. An audit log records changes to observations, layer publications, and report snapshots.

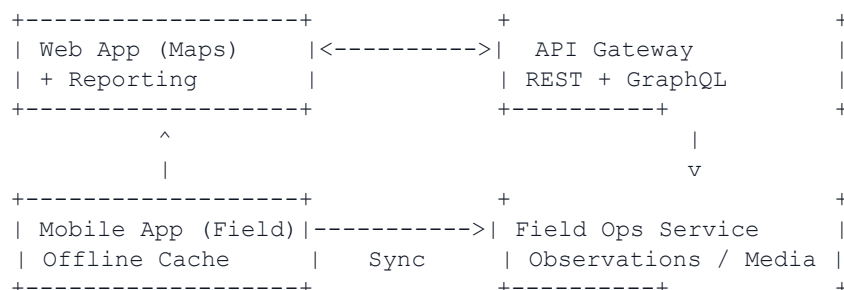
6.6 Data lineage & versioning (what changed, when, why)

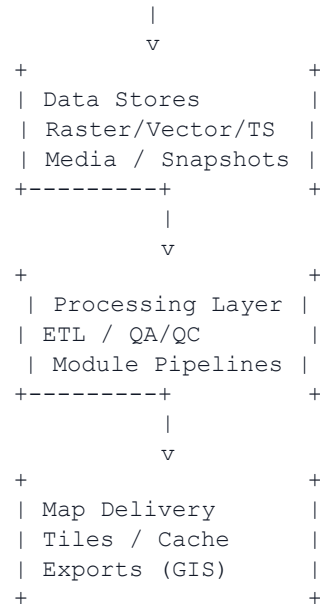
V1 requires lineage metadata for datasets and derived layers. Reports and share links reference reproducible snapshots, and the system surfaces deltas since last publish.

7. Technical Architecture (V1)

7.1 High-level architecture (frontend, backend, storage, compute)

V1 uses a map-centric frontend, an API gateway (REST and GraphQL), a field operations service for observations and media, a processing layer for ETL/QA/QC and module pipelines, and a map delivery layer for tile serving, caching, and exports.





7.2 APIs (endpoints, auth, rate limits)

V1 exposes a partner-usable API surface with tenant isolation and role-based access. Auth supports user sessions and API keys for server-to-server access. Core endpoints include terrain tiles/derivatives, module overlays and summaries, observations/media, reports/snapshots, and (enterprise) webhooks.

7.3 Geospatial stack (tiling, raster/vector, caching)

V1 is tile-first for performance and offline operation. It supports raster and vector layers, caches frequently accessed tiles for active zones, and produces exports compatible with common GIS tools.

7.4 Deployment model (cloud + edge/offline considerations)

V1 supports cloud-first deployment with offline-first clients. Enterprise deployments may enable edge processing for sensitive data and constrained connectivity. Connectivity enhancements (e.g., Starlink/LEO backhaul, mesh, LoRa) are optional and deployment-driven.

7.5 Observations ingestion format (schema example JSON)

```

{
  "schema_version": "1.0",
  "observation_id": "obs_01J4XK9Y9Q3A2K8H0V7M",
  "org_id": "org_example",
  "created_at": "2026-03-03T18:12:44Z",
  "created_by": {"user_id": "usr_123", "role": "field_tech"},
  "location": {"type": "Point", "coordinates": [-111.6543, 40.5812], "horizontal_accuracy_m": 6.5},
  "observation_type": "avalanche_snowpack",
  "tags": ["wind_slab", "recent_loading"],
  "confidence": "medium",
  "summary": "Shooting cracks on E aspects near ridgeline; recent wind transport.",
  "details": {"media": [{"media_id": "med_456", "type": "photo", "sha256": "..."}]},
  "processing": {"qa_flags": [], "review_status": "pending"}
}

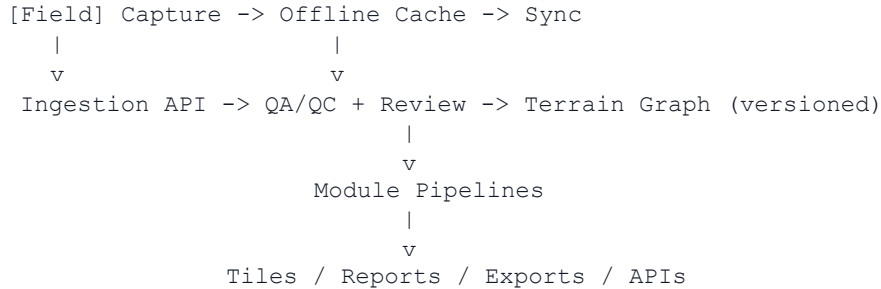
```



7.6 Security, privacy, and data governance

V1 governance includes encryption in transit and at rest, tenant isolation, role-based access control, retention controls for media and observations, and audit logs for high-impact actions.

Cross-organization sharing requires explicit agreements and enforceable permissions.



8. Safety & Operational Use

8.1 What TerraSatch is / is not (decision support, not a guarantee)

TerraSatch is a terrain intelligence and decision-support platform. It is not a guarantee of safety, and it does not replace professional judgment, training, or official warnings and advisories.

8.2 Recommended operational protocols (verification, uncertainty)

- Two-source rule for high-risk calls: do not rely on a single overlay or a single observation.
- Uncertainty-first communication: every overlay includes confidence and evidence pointers.
- Reviewer gates for publishing organization-wide layers and high-severity alerts.
- After-action packages: snapshot + evidence + audit log to support learning and defensible documentation.

8.3 Communication standards and terminology alignment

V1 supports plain-English capture mapped to structured fields. Organizations can enable standardized vocabularies and templates to align communication across shifts and teams.

9. Metrics & Validation

9.1 Model performance metrics (precision/recall, calibration, ROC where relevant)

- Classification outputs: precision/recall/F1; confusion matrices by terrain band (elevation/aspect/slope).
- Segmentation outputs: IoU and boundary error for mapped features and change detection.
- Probabilistic scores: calibration curves, Brier score, ROC-AUC where labels exist; threshold tradeoff curves.

9.2 Forecast verification and backtesting strategy



V1 emphasizes verification against historical outcomes and permitted event inventories. Verification reports are stored, versioned, and linked to the model and dataset versions used.

9.3 Field feedback loops (ground truth capture)

- Confirm/contradict annotations for published layers.
- Reviewer workflow for corrections and model improvement queues.
- Operational health metrics: observation completeness, QA flags, review latency, sync success rate.

9.4 Business/operational impact metrics (time saved, incidents reduced, etc.)

- Time to produce daily briefings and incident packages.
- Reduction in rework and “stale map” failures.
- Adoption metrics (active users, zones viewed, exports, share links used).
- Safety outcomes are not claimed as direct causality; V1 focuses on measurable workflow improvements that support safer operations.

10. Market & Use Cases

10.1 Avalanche operations & forecasting

Daily briefings, hazard mapping, and observation-driven overlays with AvyTS.

10.2 Transportation corridors & canyon road safety

Multi-hazard watch zones for avalanche, rockfall, and hydrology exposure; incident packaging.

10.3 Utilities & critical infrastructure

Segment-level exposure and inspection workflows with evidence packaging (InfraTS).

10.4 Land management & emergency response

Situational awareness, offline-first mapping, and shareable briefs for planning.

10.5 Outdoor industry partners

Partner-ready briefings and risk communication artifacts with scoped access and governance.

11. Business Model (V1)

11.1 Pricing approach (tiers, seats, org licensing)

TerraSatch offers tiered plans (see pricing page) typically aligned to team size, deployment needs, API access, and support/SLA requirements. Enterprise agreements may include webhooks, custom integrations, and data residency options.

11.2 Professional services (mapping, setup, training)

Professional services include zone setup, data onboarding (existing shapefiles, historical observations), workflow training, and integration implementation (exports, APIs, webhooks) where



contracted.

11.3 Partnerships (data providers, agencies, resorts)

Partnership models include operational pilots, data exchange agreements, and technical integrations. TerraSatch supports partner systems via documented APIs and export formats and does not require partners to replace existing GIS platforms.

12. Competitive Landscape

12.1 Market categories and what users buy today

Most organizations assemble terrain decision workflows from general GIS platforms, hazard advisory products, remote sensing vendors, and point-solution field apps. TerraSatch V1 focuses on being the operational layer that integrates these inputs rather than replacing them.

General GIS platforms (desktop + enterprise GIS)

- Strengths: mature data management and analysis.
- Gaps for field ops: offline-first capture, daily briefings, consistent audit trails.
- TerraSatch stance: integrate via exports/APIs and operate as the day-to-day field layer.

Hazard advisory / forecasting products

- Strengths: domain expertise and advisory authority.
- Gaps: limited fusion with high-resolution terrain + field observation pipelines with lineage.
- TerraSatch stance: treat advisories as authoritative inputs where applicable; add terrain-aware operational context and auditability.

Remote sensing vendors / drone deliverables

- Strengths: high-quality capture and processing deliverables.
- Gaps: stop at data delivery rather than operational decision workflows.
- TerraSatch stance: ingest standard formats and convert to ops-ready layers, alerts, and reports.

12.2 Differentiation: integrated terrain graph + field-to-model loop

- Shared Terrain Graph across hazards with consistent semantics and lineage.
- Offline-first capture and sync designed for remote field operations.
- Evidence-linked reporting and reproducible snapshots.
- API-first integration (REST + GraphQL) with enterprise webhooks.

12.3 Capability matrix (high-level)

Capability	GIS	Advisory	RS/Drone	Point App	TerraSatch
Offline capture + sync	Varies	Limited	Not core	Yes	Yes
Multi-hazard terrain model	Manual	No	No	No	Yes
Audit trail + lineage	Complex	Limited	Partial	Limited	Yes



LiDAR/drone → ops outputs	Analyst	No	Deliver	No	Ops-ready
Briefing + incident packs	Manual	Partial	No	Partial	Yes
APIs + webhooks	Yes	Varies	Varies	Limited	Yes

13. Roadmap

13.1 Phase 0 (current): foundation + pilot baseline

- TerraGrid baseline: DEM/LiDAR ingestion, derived layers, tiling and caching.
- Observation capture baseline: schema, media upload, offline storage, conflict-safe sync.
- AvyTS pilot outputs: zone overlays, storm-cycle timeline baseline, weak-layer tracking (review gated).
- Operational reporting v0: daily briefing template and share links tied to snapshots.
- Governance baseline: RBAC, tenant isolation, audit logs for publish/approve/share actions.

13.2 Phase 1 (V1): operational release

- Web app: layer control, time slider, zone summaries, exports, reporting.
- Mobile: offline-first capture, resumable uploads, deterministic sync conflict handling.
- APIs: documented terrain, AvyTS hazard, and field operations endpoints; rate limits.
- Terrain Graph v1: versioned layers, deltas, reproducible snapshots.
- Enterprise hooks: webhooks, integration templates, optional on-prem playbook (where contracted).
- Validation package: backtesting baseline and operational impact metrics for pilot orgs.

13.3 Phase 2 (V2): cross-module depth + automation

- GeoTS corridor pilots with event inventories.
- HydroTS basin pilots with targeted calibration where telemetry exists.
- PyroTS burn mapping and risk windows with clearer uncertainty.
- InfraTS integrations into work-order systems (where contracted).
- MLOps: evaluation dashboards and safe rollout/rollback tooling.

13.4 Phase 3 (scale): multi-region operations

- Repeatable ingestion + QA/QC across regions.
- Partner onboarding playbooks and reference integrations.
- Reliability engineering: SLIs/SLOs for tile delivery, ingestion latency, sync success.

14. Risks & Mitigations

- Technical: data inconsistency, model miscalibration, offline sync conflicts (mitigate with QA/QC, calibration, append-only logs, integrity checks, reviewer gates).



- Operational: workflow mismatch and over-reliance (mitigate with pilots, training, explicit uncertainty, and safety protocols).
- Legal/safety: liability exposure and privacy (mitigate with decision-support framing, auditability, encryption, RBAC, and retention controls).
- Market: procurement cycles and trust barriers (mitigate with low-friction pilots and integration-first adoption).

15. Team & Governance

15.1 Team

Keaton McCune — Founder and Technical Lead at TerraSatch. Entrepreneur building TerraSatch and SherpAI, focused on field-ready terrain intelligence, geospatial systems, and auditable decision-support workflows. More background: <https://linkedin.com/in/keaton-m>

Ericka Downs — Co-founder and Operations Lead at TerraSatch. Current student at the University of Utah majoring in Geoengineering, leading field operations, partner deployments, and workflow validation to ensure TerraSatch works under real-world constraints.

SherpAI — TerraSatch's platform agent for field and ops support. Helps structure observations, draft briefings from selected evidence, and summarize what changed between published snapshots, while staying within decision-support boundaries (no go/no-go decisions).

15.2 Governance

- Safety review for hazard-related UX and publishing workflows.
- Model change control: versioning, evaluation gating, rollout/rollback.
- Operator advisory input for validation and terminology alignment.

16. Appendix

16.1 Data source examples (public + partner-by-agreement)

- Public/common sources: satellite DEMs, public LiDAR where available, weather station networks, public incident records where available.
- Partner sources (by agreement): patrol logs and control work records, corridor inventories, asset inventories, inspection records, private drone missions.

16.2 Sample schema (report metadata)

```
{
  "schema_version": "1.0",
  "report_id": "rpt_01J4XKQ2Z7G6Y9M1",
  "org_id": "org_example",
  "type": "daily_briefing",
  "created_at": "2026-03-03T16:45:00Z",
  "time_window": {"start": "2026-03-02T00:00:00Z", "end": "2026-03-03T16:45:00Z"},
  "included_layers": [{"layer": "TerraGrid.slope", "version": "vYYYY.MM.DD.N"}],
  "exports": [{"type": "pdf", "uri": "storage://.../rpt.pdf"}], "audit":
  {"generated_from_snapshot": true, "snapshot_id": "snap_..."}
}
```

16.3 Glossary



Term	Definition
AOI (Area of Interest)	A defined geographic area used for analysis, map rendering, caching, and reporting.
API (Application Programming Interface)	A programmatic interface used to request/push data between TerraSatch and other systems.
Audit Log	A record of high-impact actions (publish, approve, edit, share) including actor, timestamp, and object changed.
AvyTS (AvyTerraSatch)	TerraSatch module focused on avalanche and snowpack intelligence (terrain + weather + observations).
Backtesting	Evaluating model outputs against historical periods and known outcomes to estimate performance.
Calibration	How well probabilistic scores match reality (e.g., predicted likelihood aligns with observed frequency).
Change Detection	Identifying meaningful differences between datasets across time (surface change, vegetation change, burn scar, etc.).
Confidence	A communicated level of certainty tied to data coverage/quality and model validation status.
Curvature	A terrain derivative describing how slope changes (convex/concave features).
DEM (Digital Elevation Model)	A gridded elevation surface used to derive slope, aspect, flow routing, and other features.
Derived Layers (Terrain Derivatives)	Computed layers from elevation/surfaces such as slope, aspect, curvature, roughness, flow accumulation.
Edge / Offline-first	Design where field devices remain usable without connectivity and synchronize when back online.
ETL (Extract, Transform, Load)	Pipeline steps to ingest raw data, normalize/QA it, and store it for analysis and delivery.
Field Observation	A structured record captured in the field (GPS location, timestamp, tags, narrative, photos, measurements).



Flow Accumulation	Hydrology derivative estimating how water concentrates and routes across terrain.
GeoTS	TerraSatch module for rockfall/slope-stability screening and geologic terrain context (not engineering certification).
HydroTS	TerraSatch module for watershed/runoff/flood context and event awareness.
InfraTS	TerraSatch module for infrastructure inspection workflows and hazard exposure mapping.
IoU (Intersection over Union)	Segmentation metric measuring overlap between predicted and reference mapped areas.
LiDAR (Light Detection and Ranging)	Remote sensing method producing dense 3D point clouds used for high-resolution terrain/surface modeling.
Lineage	Traceability of outputs back to inputs, processing steps, versions, and review status.
ML (Machine Learning)	Models that learn patterns from data; in TerraSatch V1 used where outputs are explainable and auditable.
Multi-hazard	Combining multiple hazard domains (avalanche, wildfire, flood, rockfall) into a shared terrain framework.
Orthomosaic	Geometrically corrected stitched aerial imagery (often drone-derived) used as a map layer.
Point Cloud	A collection of 3D points (LiDAR/photogrammetry) representing surfaces and objects.
Precision / Recall	Classification metrics. Precision penalizes false positives; recall penalizes false negatives.
PyroTS	TerraSatch module for wildfire/fuels/risk context and incident planning layers.
Raster / Vector	Raster: pixel/grid data (DEMs, satellite indices). Vector: points/lines/polygons (routes, boundaries, assets).
RBAC (Role-Based Access Control)	Access permissions based on user roles (admin, forecaster, field operator, viewer).



Remote Sensing Indices	Computed values from satellite imagery used for vegetation/fuels and burn severity context (e.g., NDVI, NBR).
Review Gate	A workflow requirement where an authorized reviewer approves before publishing a layer/report broadly.
Roughness	Terrain derivative indicating surface variability; used as a contextual feature in multiple modules.
Snapshot	A reproducible state of layers/observations/time window used for reporting and auditability.
Slope / Aspect	Slope is steepness; aspect is compass direction the slope faces. Core terrain derivatives for AvyTS.
SNOTEL	NRCS Snow Telemetry network reporting snowpack and meteorological measurements in mountain watersheds.
Terrain Graph	TerraSatch's versioned representation linking terrain tiles, derivatives, observations, outputs, events, and assets.
TerraGrid	TerraSatch's unified base terrain model and derivative stack used across all modules.
Tiling	Breaking geospatial layers into tiles for fast map rendering, caching, and offline use.
Uncertainty	Communicated limits of confidence due to incomplete data, rapid change, or unvalidated assumptions.
Watch Zone	A mapped area flagged for attention based on thresholds or conditions, with evidence and confidence indicators.

16.0 References

TerraSatch

1. TerraSatch main site: <https://terrasatch.com>
2. TerraSatch data portal: <https://data.terrasatch.com>

Avalanche operations and standards

3. Avalanche.org (U.S. forecast + education hub; A3 + NAC partnership): <https://avalanche.org/>
4. Avalanche.org About (partnership description): <https://avalanche.org/about/>



5. American Avalanche Association (A3): <https://www.americanavalancheassociation.org/>
6. SWAG Guidelines landing page (overview of SWAG): <https://www.americanavalancheassociation.org/swag>
7. SWAG (PDF version hosted online; may not be the latest edition):
https://static1.squarespace.com/static/59d2a0f0e9bdf20d6d654b7/t/5a1af2a5652dea2e1a5ea055/1511715529879/AAA_SWAG_Web%2B2.pdf
8. National Avalanche Forecast Platform (AFP) overview (NAC): <https://avalanche.org/national-avalanche-center/>

Weather and alerts (common operational sources)

9. National Weather Service (NWS): <https://www.weather.gov/>
10. NWS Active Alerts: <https://www.weather.gov/alerts>

Terrain and elevation (DEMs and LiDAR access)

11. USGS 3D Elevation Program (3DEP): <https://www.usgs.gov/3d-elevation-program>
12. USGS 3DEP products and services: <https://www.usgs.gov/3d-elevation-program/about-3dep-products-services>
13. USGS The National Map 3DEP DEM viewer: <https://apps.nationalmap.gov/3depdem/>
14. OpenTopography (public topographic + LiDAR access): <https://opentopography.org/>
15. OpenTopography “Find Data”: <https://opentopography.org/data>

16.4.5 Hydrology and watershed context

16. USGS Water Data for the Nation: <https://waterdata.usgs.gov/>
17. USGS NWIS Mapper tool: <https://apps.usgs.gov/nwismapper/>
18. USGS National Water Dashboard: <https://dashboard.waterdata.usgs.gov/>
19. NRCS Snow and Water Interactive Map (SNOTEL/snow survey access):
<https://www.nrcs.usda.gov/resources/data-and-reports/snow-and-water-interactive-map>
20. NRCS SNOTEL Historic Data: <https://wcc.sc.egov.usda.gov/nwcc/tabget>

Wildfire, fuels, and burn severity (planning context)

21. National Interagency Fire Center (NIFC): <https://www.nifc.gov/>
22. NIFC Incident Information: <https://www.nifc.gov/nicc/incident-information>
23. RAWs (Remote Automatic Weather Stations): <https://raws.nifc.gov/>
24. LANDFIRE data portal: <https://www.landfire.gov/data>
25. LANDFIRE Fuel products overview: <https://www.landfire.gov/fuel>

Note: Items above are “common sources.” TerraSatch only represents partner or proprietary datasets as sources when explicitly provided under agreement e.g. through a provider like [Utah Avalanche Center](#)

